Experiment Progress Report

Interleaved Training v3 & Sequential Dead Neuron Reinitialisation

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This report covers results from two new experiments on catastrophic forgetting and capacity loss in deep RL. Both were trained for 2 M total environment steps. Section 1 gives the baseline context, Sections 2 and 3 cover each experiment in detail, and Section 4 contains the plots.

Context: Existing Baselines

One framing point worth flagging upfront. The sequential and interleaved conditions cannot be directly compared on the Pong column. Sequential conditions transfer from a fully-trained Pong agent (+9.1 reward) and the question is how much performance gets *retained*. The interleaved conditions start from scratch and the question is how much Pong performance gets *developed* under joint training constraints. This distinction matters when reading Table 2.

Condition	Pong start	Pong end (chimera)	Dead neurons (end)
No freeze (full fine-tune)	+9.1	−20.8	87.3%
Freeze conv only	+9.1	−14.0	83.6%
Freeze conv + fc_repr	+9.1	+7.2	87.5%

Table 1: Sequential transfer baselines. All start from a Pong DQN trained for 2 M steps. Dead neurons are measured on Breakout frames at the end of 2 M Breakout steps.

The takeaway from the baselines is that freezing fc_repr alone is enough to preserve Pong performance almost perfectly. Once fc_repr is allowed to update, Pong collapses quickly even when the conv layers stay frozen.

Experiment 1: Sequential + Dead Neuron Reinitialisation

Purpose

The hypothesis was straightforward. After Pong training leaves 76% of fc_repr neurons dead, reinitialising those neurons before starting Breakout might give the network more capacity for the new task and reduce forgetting. The idea was that Breakout would be using otherwise idle parameters rather than overwriting the ones Pong actually used.

Conduct

After 2 M Pong steps, we identified dead neurons in fc_repr using the criterion that post-ReLU activation must equal zero in over 95% of 1 000 sampled Pong frames. 388 out of 512 neurons (75.8%) met this threshold. All 388 were reinitialised using orthogonal initialisation (gain $= \sqrt{2}$, biases to zero), which is the same scheme used at network creation. The Adam optimiser was also recreated to clear stale momentum on those weights. Breakout training then ran for 2 M steps under otherwise identical conditions to the no-freeze baseline.

Results

Event / Step	Pong (chimera)	Breakout	Dead neurons	CKA drift
After Pong training (pre-reinit)	+9.1	—	75.8%	—
After reinit, step 0 (pre-Breakout)	**−20.9**	0.0	**33.0%**	0.065
200k Breakout steps	−21.0	2.1	47.5%	0.018
1M Breakout steps	−21.0	3.4	63.1%	0.018
2M Breakout steps (final)	**−21.0**	**3.5**	**77.5%**	**0.025**

Table 2: Sequential reinit metrics. The bold step-0 row is after reinit but before any Breakout training. Pong reward is the chimera evaluation, which combines the current backbone weights with the original frozen Pong head. Dead neurons are measured on Breakout frames.

Interpretation

The forgetting happened at reinit, not during Breakout training.

Pong reward collapses to −20.9 at step 0, before a single Breakout gradient step. The forgetting was not caused by Breakout overwriting Pong features. It was caused by the reinitialisation itself.

The reason involves a subtle co-adaptation issue. After 2 M Pong steps, 388 fc_repr neurons are dead, meaning their post-ReLU output is consistently zero on Pong frames. The Pong output head maps the full 512-dimensional representation to Q-values, but since 388 of those dimensions are always zero, the head has effectively learned to work in a 124-dimensional subspace. The weight columns corresponding to the dead neurons are essentially arbitrary. They were trained with those neurons silent, so their values never mattered.

After reinitialisation, those 388 neurons get new random orthogonal weights and start producing large non-zero activations on Pong frames. The Pong head is now receiving 388 unexpected inputs from dimensions it had always treated as silent. The Q-value estimates are immediately corrupted, not by any Breakout interference, but by the reinit violating the implicit agreement between the backbone and the head.

The core insight here is that **dead neurons are not just wasted capacity. They are locked-in silence that the output layer depends on.** Resetting them without also adapting the downstream layer is destructive regardless of what comes next. This is precisely why Continual Backpropagation (Dohare et al. 2024) works differently. It reinitialises neurons gradually throughout training so the rest of the network can adjust incrementally rather than absorbing a sudden structural shock all at once.

Dead neurons re-accumulate at the same rate regardless.

After reinit, the dead fraction dropped from 76% to 33% on Breakout frames. By 2 M Breakout steps it had climbed back to 77.5%, which is actually higher than before the reinit. The rate of accumulation is approximately 22 percentage points per 1 M steps, which closely matches the rate seen during Pong training from scratch at approximately 38pp per 1 M steps. This confirms that dead neuron accumulation is not a Pong-specific problem or a consequence of transfer. It is a general property of deep RL training with ReLU activations and Adam, and it happens at roughly the same pace regardless of starting conditions.

It is also worth noting that the dead neuron fractions here are measured on Breakout frames rather than Pong frames. A neuron that is alive on Pong states may well be dead on Breakout states and vice versa. The measure is always relative to the input distribution, which is something to bear in mind when comparing across conditions.

Breakout learning is unaffected either way.

Breakout reward reaches 3.5 at 2 M steps, matching the no-freeze baseline exactly. The reinit neither accelerated nor harmed new task acquisition. Either the 124 Pong-alive neurons were sufficient for Breakout, or the reinited neurons simply died again before making a meaningful contribution.

See Figure 1 for the full timeline.

Experiment 2: Interleaved Training v3 (Episode-Level, Joint Buffer)

Purpose

The previous interleaved design (v2) used step-level switching every 1 000 steps with two separate replay buffers. It failed completely. Pong collapsed to −21 and Breakout stalled at 0.4, so both tasks were lost. This experiment implements the redesign you described: episode-level switching with a single joint replay buffer, to see if that resolves the failure.

Conduct

Two changes were made relative to v2:

1. **Episode-level game switching.** The game only switches at natural episode boundaries rather than mid-episode. At each boundary, the game with fewer accumulated steps is selected, keeping both games balanced in step count throughout training. This matters because Pong episodes average around 200 steps while Breakout episodes average around 8 steps. Strict alternation by episode would give Pong 96% of the total steps, so step-based balancing was necessary.
2. **Single joint replay buffer.** All transitions from both games are stored in one buffer, each tagged with a game ID. Every training batch contains transitions from both games. Pong transitions are routed through the Pong output head and Breakout transitions through the Breakout head. The losses are summed and a single backward pass sends gradients from both games through the shared backbone simultaneously on every update.

The architecture is a two-head network with a shared conv and fc_repr backbone (512-dim) and separate per-game output heads. It was trained entirely from scratch with no Pong checkpoint used. The budget was 1 M steps per game (2 M total), with a learning rate of 10^{-4}, 100k buffer capacity, and batch size 32.

The reason v2 likely failed is the separate buffers. Each training batch was 100% from one game, so the backbone alternated between Pong-only and Breakout-only gradient updates that largely cancelled each other. The joint buffer fixes this because every update is a mixed-game signal and the backbone never has to serve a single game in isolation.

Results

Interpretation

Finding 1: The dead neuron trajectory is non-monotonic.

Dead neuron fraction drops from 47% to 34% over the first 600k steps, then rises to 62% by

Total steps	Pong reward	Breakout reward	Dead neurons	CKA drift
0	0.0	0.0	47.1%	0.010
300k	−13.4	3.5	45.1%	0.046
600k	−9.2	2.9	**34.0%**	0.076
700k	−7.5	3.2	37.3%	0.040
1.7M	−4.7	2.9	52.5%	0.065
2M (final)	**−6.8**	**3.2**	**62.1%**	**0.039**

Table 3: Interleaved v3 metrics at key checkpoints. Pong and Breakout rewards are online evaluations during joint training. Both games start from scratch, so the Pong reward of −6.8 is developed through joint training rather than retained from a pretrained agent.

2 M steps. This U-shape does not appear in any other condition and is the most mechanistically novel result from these experiments.

At the start of training, 47% of neurons are already dead, which is typical for a randomly initialised ReLU network with zero biases. What is unusual is that joint training then reduces the dead fraction before the usual rise sets in. The reason is that gradient signals from both games reach the backbone simultaneously on every update. A neuron that Pong training alone would kill, because Pong does not need the feature it responds to, may be kept alive by Breakout gradients that do need it, and vice versa. The two-game signal acts as a diversity pressure, preventing the backbone from collapsing into a single game's representational subspace.

After around 600k steps, both games' Q-functions are sufficiently mature that the backbone starts refining toward more specialised features. Neurons not needed by either game begin to die and the fraction rises again. The fact that it only reaches 62% rather than the 87% seen in sequential conditions suggests the joint pressure is still active, just weaker than in the early phase.

The practical implication is that the plasticity benefit of joint training is strongest in the first half of training. Whether continuing beyond that point gradually erodes it further is worth tracking over longer runs.

Finding 2: Breakout is learned much faster under joint training.

Breakout reward reaches around 3.5 by 300–400k total steps, which corresponds to roughly 150–200k Breakout-specific steps. The standalone Breakout DQN takes the full 2 M dedicated steps to reach 3.86. This is roughly an $8\times$ speedup in Breakout-step terms.

The most likely explanation is shared visual structure. Both Pong and Breakout involve a moving ball against a structured background. The features useful for Pong, such as detecting ball position, estimating velocity, and tracking relative positions, are also exactly what Breakout needs. Training on Pong from the first step effectively pre-builds these features so Breakout can use them immediately rather than having to learn them from scratch.

Finding 3: Pong stabilises around −7 and stays there.

Pong reward reaches around −7 to −5 by 700k steps and remains at that level for the remaining 1.3M steps of training. This is not ongoing interference or instability. It is a settled plateau where the two-head network has found a stable shared representation that serves both tasks adequately. The variance across evaluation points reflects noise in 10-episode means rather than any real shift in the underlying policy.

The plateau at −7 rather than the standalone Pong optimum of +7.5 reflects the multitask cost. The backbone must serve both games, and the best shared representation is not the best

Pong-only representation. The gap of roughly 14 reward points between them is the price of joint training without dedicated Pong capacity.

See Figures 2 and 3 for the full trajectories.

Summary

Condition	Pong final	Context	Breakout final	Dead (final)
No freeze (sequential)	−20.8	retained from +9.1	3.9	87.3%
Freeze conv only (sequential)	−14.0	retained from +9.1	3.7	83.6%
Freeze all (sequential)	+7.2	retained from +9.1	—	87.5%
Sequential + reinit (new)	−21.0	lost at reinit step	3.5	77.5%
Interleaved v2 (step-level)	−21.0	developed from scratch	0.4	92.2%
Interleaved v3 (episode + joint)	−6.8	developed from scratch	3.2	62.1%
Standalone Pong DQN	+7.5	2M dedicated steps	—	—
Standalone Breakout DQN	—	2M dedicated steps	3.86	—

Table 4: All conditions. The Context column clarifies whether sequential Pong values measure retention from a pretrained agent or development from scratch under joint training, as these are fundamentally different quantities.

Three things stand out from this set of results. First, dead neurons are not inert. They are structurally co-adapted with the downstream layer, and resetting them in one shot causes immediate damage rather than restoring capacity. Second, joint training with a shared buffer produces a non-monotonic plasticity curve where the backbone initially becomes more plastic under joint supervision before gradually specialising. Third, episode-level switching with a joint buffer (v3) succeeds at both tasks while step-level switching with separate buffers (v2) failed both. The joint buffer appears to be the critical design variable.

Plots

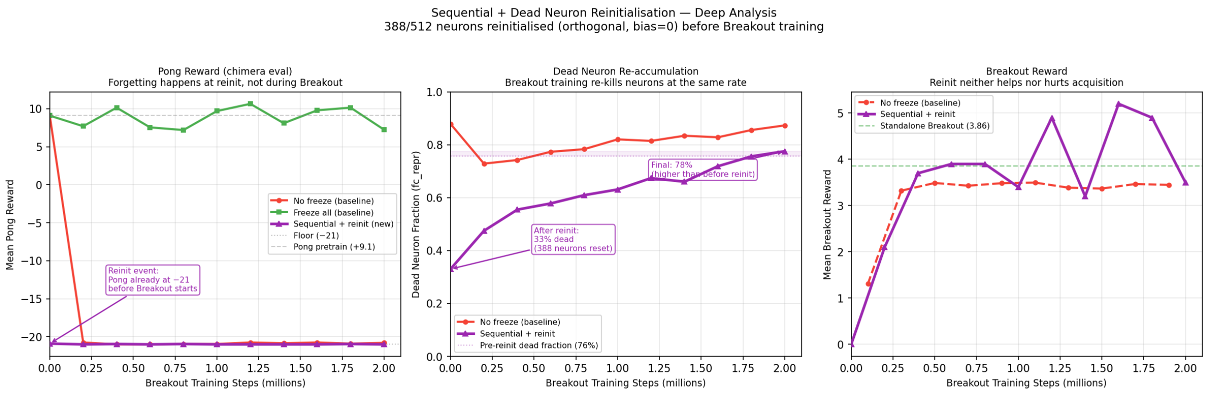

Figure 1: **Sequential + Dead Neuron Reinitialisation.** Left: Pong reward is already at floor before any Breakout training, showing that forgetting occurs at reinit rather than during Breakout. Centre: dead neurons drop from 76% to 33% at reinit then climb back to 77.5% over 2M Breakout steps. Right: Breakout reward is identical to the no-freeze baseline, confirming that reinit neither helps nor hurts new task acquisition.

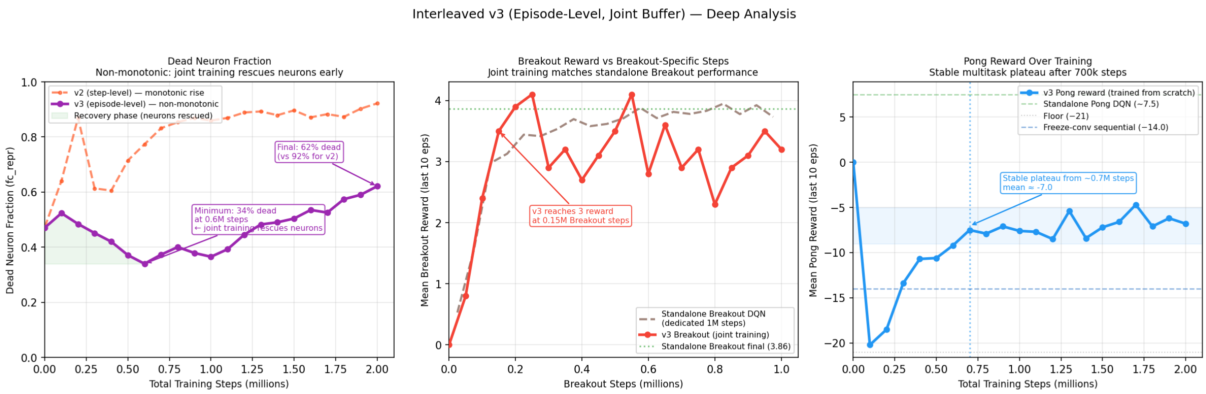

Figure 2: **Interleaved v3 deep dive.** Left: dead neuron fraction drops during the first 600k steps as joint gradients rescue neurons, then rises during re-specialisation. Centre: v3 Breakout reward plotted against Breakout-specific steps, reaching standalone Breakout performance in around 150k steps compared to 2M for the dedicated DQN. Right: Pong reward stabilises around -7 from 700k steps onwards.

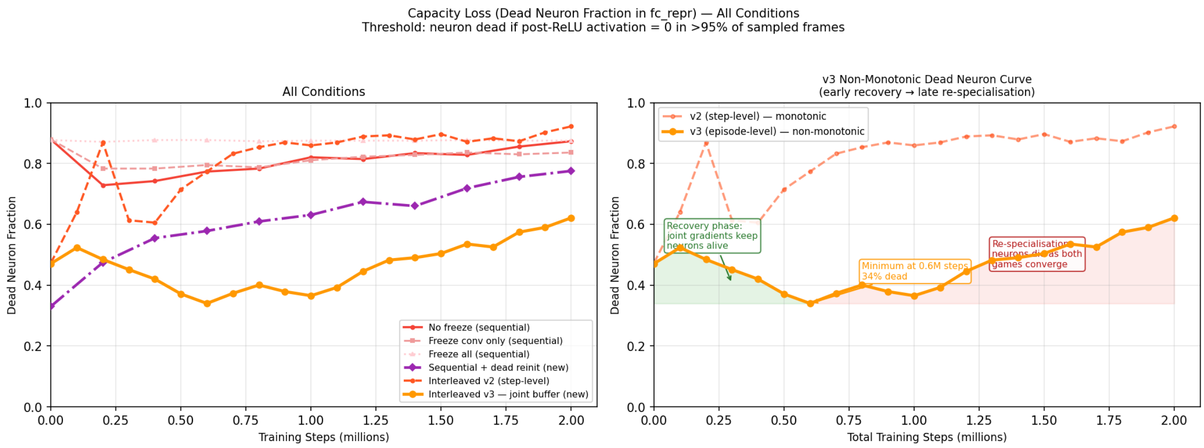

Figure 3: **Capacity loss across all conditions.** Left: all six conditions together. Right: v3 non-monotonic curve annotated with the recovery phase in green and the re-specialisation phase in red. v3 ends at 62% dead, which is 30 percentage points below both the sequential no-freeze baseline and v2.

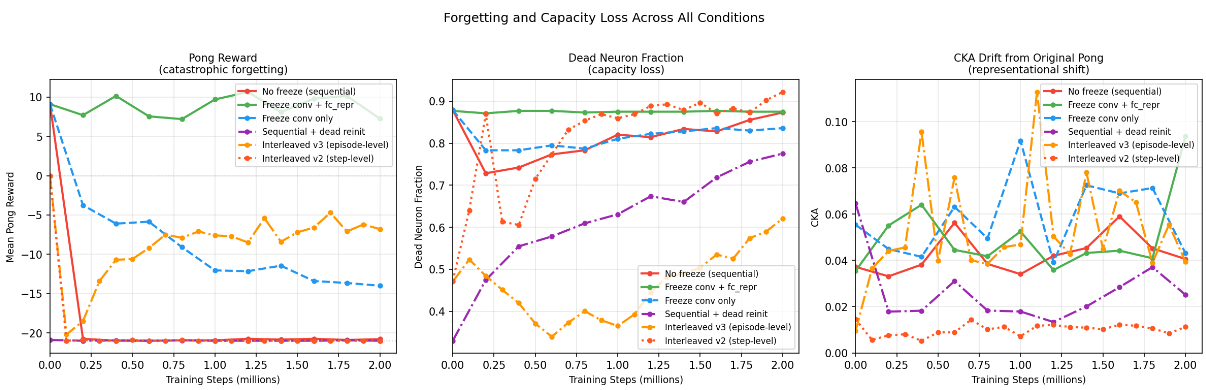

Figure 4: **All conditions overview.** Pong reward, dead neurons, and CKA drift across all six conditions. The x-axis is Breakout steps for sequential conditions and total steps for interleaved conditions. CKA values use random Pong frames as the probe set and should be read as indicative rather than definitive given the cross-game distribution mismatch.